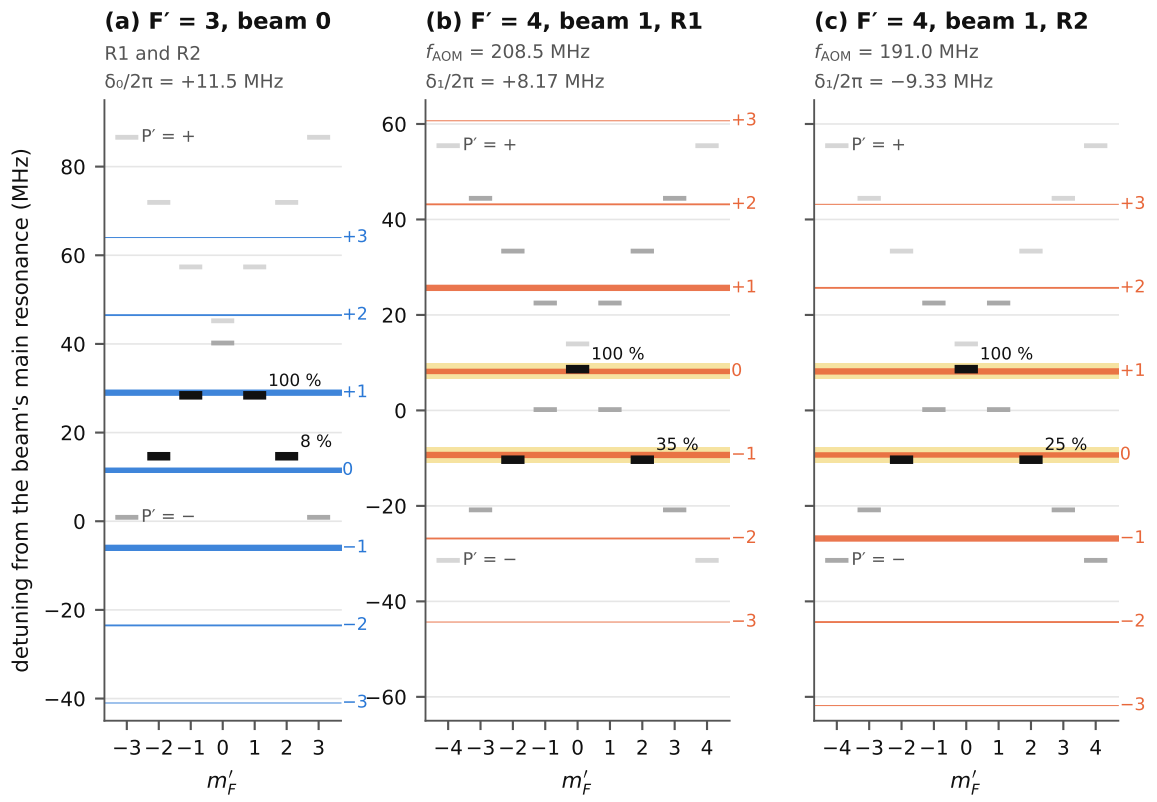

(d) why both AOM frequencies work

$f_{AOM}(R1) - f_{AOM}(R2) = 208.5 - 191.0\ \text{MHz} = 17.5\ \text{MHz} = f_{EOM}$

→ the beam-1 lines lie at the same frequencies in R1 and R2; only their order n shifts by one

→ carrier $J_0^2(1.5) = 0.26 \approx$ first order $J_1^2(1.5) = 0.31$, so $m'_F = 0$ and ±2 (yellow) get almost the same power

→ the bottleneck $|F = 3, m_F = \pm 3\rangle$ reaches F′ = 4 only via $m'_F = \pm 3, \pm 4$; R2's −1 order at −26.8 MHz pumps it 15 % faster than R1, hence 99.85 % (R2) against 99.82 % (R1)

bars: excited sublevels, same m'_F stacked; black = driven; line width ∝ sideband power

%: excitation rate relative to the most strongly driven sublevel of the same beam; black if ≥ 5 %

rate = weak-field estimate, Σ over J = 2 sublevels and sidebands of $|M|^2$ × sideband power × Lorentzian (width Γ)